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SIMATS SCHOOL OF ENGINEERING

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B.E./ B.Tech DEGREE EXAMINATION - AUG 2026
MODEL PRACTICAL EXAMINATIONS
CSA51 - CRYPTOGRAPHY AND NETWORK SECURITY
SET 7

1. The **Hill Cipher** uses matrix multiplication for encryption. Given the plaintext "ACT" and a 3x3 Key Matrix

$$K = \begin{bmatrix} 6 & 24 & 1 \\ 13 & 16 & 10 \\ 20 & 17 & 15 \end{bmatrix}$$

- a) Convert the plaintext "ACT" into its numerical vector representation.
b) Compute the ciphertext vector using $C = (K \cdot P) \pmod{26}$ Show matrix multiplication
2. To increase security, a sender applies a **Double Transposition Cipher** to the message "MEET ME AT NOON"
- A) First, apply a simple **Rail Fence Cipher** with a depth of 2
B) Take the output of the Rail Fence and apply a **Columnar Transposition** using the key "3142"
3. Write a C program for DSA, because the value of k is generated for each signature, even if the same message is signed twice on different occasions, the signatures will differ. This is not true of RSA signatures. Write a C program for implication of this difference
4. Write a C program for SHA-3 option with a block size of 1024 bits and assume that each of the lanes in the first message block (P0) has at least one nonzero bit. To start, all of the lanes in the internal state matrix that correspond to the capacity portion of the initial state are all zeros. Show how long it will take before all of these lanes have at least one nonzero bit.

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CSA 5116 - CNS

MODEL PRACTICAL EXAMINATIONS

1. Aim:-

To Implement the Hill Cipher program for the given key and plain text "ACT" to find the Encryption

Algorithm:-

*). Take the given plain text and make it in the form of numerical vector representation.

*). Convert each character into its numerical equivalent using A=0 to Z=25

*. Represent the plaintext as a column vector

$$P = \begin{bmatrix} 0 \\ 2 \\ 19 \end{bmatrix}$$

*. Multiply the key matrix with plaintext vector

*. Convert the resulting numbers back to corresponding alphabetic characters.

*. Display the encrypted ciphertext

*. Stop the algorithm.

pseudocode for given plain text:-

declare key [3][3]

declare plain [3],

cipher [3]

Declare i, j, sum

key = { {6, 24, 1}, {13, 16, 10}, {20, 17, 15} }

plain = {0, 2, 19} //

plain text ACT

For i=0 to 2 sum = sum + key[i][j] * plain[j]

End for

print "Cipher text : "

For i=0 to 2

print

character (cipher[i] + 'A')

END for

END.

Output:

Cipher text: POH.

a). Given:-

plain text: ACT

Its numerical representation: $\begin{bmatrix} 0 \\ 2 \\ 19 \end{bmatrix}$

b)

$$P = \begin{bmatrix} 0 \\ 2 \\ 19 \end{bmatrix}$$

$$K = \begin{bmatrix} 6 & 24 & 1 \\ 13 & 16 & 10 \\ 20 & 17 & 15 \end{bmatrix}$$

$$C = K \cdot P \pmod{26}$$

$$C = \begin{bmatrix} 6 & 24 & 1 \\ 13 & 16 & 10 \\ 20 & 17 & 15 \end{bmatrix} \begin{bmatrix} 0 \\ 2 \\ 19 \end{bmatrix} \pmod{26}$$

$$= \begin{bmatrix} 0 + 48 + 19 \\ 0 + 32 + 190 \\ 0 + 34 + 285 \end{bmatrix} \pmod{26}$$

$$= \begin{bmatrix} 15 \\ 14 \\ 7 \end{bmatrix} = \begin{bmatrix} P \\ 0 \\ 17 \end{bmatrix} = \text{POH.}$$

$$\begin{array}{r} 4 \\ 5 \times 19 \\ \hline 135 \\ 150 \\ \hline 285 \end{array}$$

2.

Aim:-

To encrypt the message "MEET ME AT NOON"
using double transposition cipher.

Algorithm:-

1. Start the program
2. Read the plaintext
3. Remove spaces from the plain text
4. Arrange the letters in rows according to the selected number of columns
5. perform the first columnar transposition.
6. Read the columns in the specified order to obtain the first intermediate ciphertext
7. Arrange the intermediate ciphertext again in a matrix.
8. perform the second columnar transposition.
9. Read the columns according to the second key.
10. Display the final ciphertext
11. stop.

Pseudo code for given plain text:-

BEGIN
Read plaintext

Remove spaces

Arrange plaintext in matrix

perform first transposition
 Read columns according to
 key 1

store result in temp

Arrange temp in matrix

perform second transposition

read columns according to

key 2

store result in temp

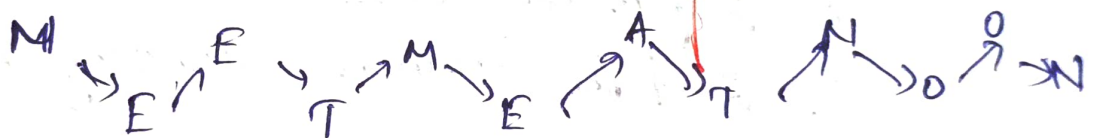
Arrange it in matrix

END the program.

Output :-

a). MEET ME AT NOON

depth = 2



ME M A N O / E T E T O N

b). Key = 3142

3	1	4	2
M	E	E	T
M	E	A	T
N	O	O	N

Cipher text : EEOTINMMN EAAO.

Aim:-

To write a C program demonstrating that DSA signatures are different when the same message is signed on different occasions, because a fresh random value k is used for every signature.

Algorithm:-

1. Start
2. Generate DSA domain parameters
3. Generate a private key and corresponding public key.
4. Enter the same message
5. Generate a fresh random k_1
6. Generate the first signature (r_1, s_1)
7. Generate another random k_2
8. Generate the second signature (r_2, s_2)
9. Compare both signatures
10. Show that the signatures are different
11. Verify both signatures using the public key
12. Stop.

pseudocode:-

BEGIN

Generate DSA parameters

Generate private and public keys

Read message

Generate random k_1

Signature $S_1 = \text{DSA_SIGN}(\text{message})$

Generate random k_2

Signature $S_2 = \text{DSA_SIGN}(\text{message})$

print = SIGNATURE 1;

print = SIGNATURE 2;

END.

4. Aim :-

To implement the SHA-3 permutation for a 1024 bits and assume that the each of the lane in the first message block (P_0) has at least one non-zero bit.

Algorithm:-

1. Start the program with the given parameters.
2. Set the first 16 lanes corresponding to the 1024 bit message block to non-zero values.
3. Set the remaining 9 capacity lanes to zero.

5. perform the SHA3 - Keccak-f[1600] perm
6. Each permutation has 24 rounds
7. Each round performs
 1. Θ (theta)
 2. ρ (rho)
 3. π (π)
 4. χ (χ)
 5. ι (ι)
8. After every round, count the number of lanes whose value is nonzero
9. Display the result.

~~Pseudo~~ Code: -

